**(b)** **(i)** Complete the Karnaugh map (K-map) for the Boolean expression:

$\mathbf{\overline{A}.\overline{B}.C + \overline{A}.B.C + A.\overline{B}.C + A.B.\overline{C}}$

| **A** \ **BC** | **00** | **01** | **11** | **10** |
|---|---|---|---|---|
| **0** | | | | |
| **1** | | | | |

[2]

**(ii)** Draw loop(s) around appropriate group(s) in the K-map to produce an optimal sum-of-products. [2]

**(iii)** Write the Boolean expression from your answer to part **b(ii)** as a simplified sum-of-products. Do **not** carry out any further simplification.

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**7** **(a)** Asymmetric encryption is a type of cryptography.

Identify **one** other type of cryptography.

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**(b)** An organisation holds two asymmetric encryption keys, which they intend to use to receive secure transmissions.

Explain how the organisation makes use of the two keys to receive a secure transmission.

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